
Engression for Causal Inference under Weak Overlap

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Abstract

Estimating causal effects under limited overlap is challenging because standard methods must extrapolate into regions with little covariate support, often leading to unstable or biased estimates. In this project, we investigate whether distributional regression via Engression can improve causal estimation in such settings. Unlike classical approaches that model only the conditional mean, Engression learns the full conditional outcome distribution, which may yield more stable counterfactual predictions under weak overlap. We consider two ways to use **Engression: as an outcome model and as a data augmentation tool**. Through our simulations, we find that replacing the outcome model with Engression consistently improves performance over classical approaches, whereas naive data augmentation often provides limited benefit. We also evaluate these ideas on empirical benchmark datasets, where augmentation improves covariate balance but does not consistently translate into better causal estimation. Overall, our results suggest that distributional regression can be a useful component for causal inference under limited overlap, but its success depends critically on how it is integrated into the estimation procedure.¹

1 Introduction and Background

Causal inference from observational data often relies on the potential outcome framework and two key identifying assumptions: (i) conditional independence (ignorability) and (ii) overlap (positivity). In practice, however, the overlap assumption is often violated, especially in high-dimensional settings or when treatment assignment is highly selective. In such cases, treated and control units do not share sufficient covariate support, forcing any estimator that requires extrapolation to rely heavily on model specification [Crump et al., 2009].

1.1 Average Treatment Effect (ATE)

In this project, our primary quantity of interest is the Average Treatment Effect (ATE), which captures the expected difference in outcomes between treated and control groups at the population level. Under the potential outcome framework, let $Y(1)$ and $Y(0)$ denote the potential outcomes under treatment and control, respectively. The ATE is defined as

$$\tau = \mathbb{E}[Y(1) - Y(0)].$$

Under the assumptions above, the ATE can be expressed as a functional of observable quantities, typically involving the conditional outcome regression $\mu_t(x) = \mathbb{E}[Y \mid X = x, T = t]$ and/or the propensity score $e(x) = \mathbb{P}(T = 1 \mid X = x)$.

1.2 Weak Overlap in Causal Inference

Formally, we use the **propensity score of treatment assignment**, defined by

$$e(x) = \Pr(T = 1 \mid X = x).$$

¹Code available at: <https://github.com/TonyLei43/stat-538-final-project>

The overlap condition requires $0 < e(x) < 1$ for almost every $X = x$. Intuitively, we want both treatment arms to be represented throughout the covariate space. However, this condition is often violated in real-world datasets because of rare treatments, selection bias, and confounding.

When overlap is weak, causal estimation becomes unstable. Propensity-based methods suffer from extreme weights, while outcome regression methods must extrapolate into regions with little or no data support, leading to potentially biased estimates. Even doubly robust estimators, which combine both approaches, can perform poorly when one nuisance component is misspecified, since lack of overlap amplifies estimation error.

Motivated by this issue, we propose using distributional regression as a more robust way to estimate treatment effects.

1.3 Distributional Regression

Instead of modeling only the conditional mean, distributional regression methods target the full conditional distribution $P_{Y|T,X}(y | t, x)$. This approach can be beneficial under weak overlap. First, by modeling the full conditional distribution, it provides information beyond the conditional mean, such as heteroskedasticity and distributional shape, which may be useful for downstream estimators. Second, learning the entire conditional density may impose additional smoothness across the outcome space, stabilizing predictions in sparsely populated regions of the covariate space. Among various approaches to distributional regression, we focus on Engression, a recent generative modeling framework designed for estimating complex conditional outcome distributions.

1.3.1 Engression

Engression, introduced by Shen [2024], is a neural-network implementation of distributional regression that optimizes the **energy score**, a strictly proper scoring rule for continuous distributions.

Let $g_\theta(t, x, \varepsilon)$ be a generator network with $\varepsilon \sim N(0, 1)$. The population loss is

$$\mathcal{L}_{\text{Eng}}(\theta) = \mathbb{E} \left[|Y - g_\theta(T, X, \varepsilon)| - \frac{1}{2} |g_\theta(T, X, \varepsilon) - g_\theta(T, X, \varepsilon')| \right],$$

where ε' is an independent copy of ε . Minimizing this loss yields a conditional distribution estimator $\hat{F}_{Y|T,X}$, from which we can generate samples and extract quantities of interest such as the conditional mean, variance, or quantiles.

1.4 Two Ways to Leverage Engression

In this project, we investigate two complementary uses of Engression in the presence of weak overlap:

- (a) **Engression as an outcome model.** We use Engression as an outcome model to estimate the conditional outcome distribution $P(Y | X, T)$ and obtain conditional mean estimates. We consider both S-learner and T-learner formulations, as well as replacing the outcome-regression component in AIPW with Engression-based predictions.
- (b) **Data augmentation.** We first train Engression on the weak-overlap sample and then identify covariate regions with poor overlap. For units in those regions, we use the trained Engression model to generate synthetic counterfactual outcomes and augment the sample with synthetic twins before applying standard causal estimators.

2 Methodology and Results

All experiments are conducted in a controlled simulation setting, with the full data-generating process described in Appendix A. In brief, covariates, propensity scores, and potential outcomes are generated following Shen [2024], after which weak overlap is deliberately induced in selected regions of the covariate space.

2.1 Generating a Weak-Overlap Dataset

Starting from the super-population described in Appendix A, we induce weak overlap through a thinning procedure. Specifically, we define a sharp propensity score

$$e_{\text{weak}}(X) = \sigma(\gamma(0.6X_1 + 0.4X_2 - 0.3X_1^2)),$$

and retain observations with probabilities that depend on their observed treatment status: treated units are more likely to be retained when $e_{\text{weak}}(X)$ is large, while control units are more likely to be retained when $e_{\text{weak}}(X)$ is small. This process creates regions of the covariate space in which one treatment arm is sparse, yielding a sample with limited overlap and more extreme propensity scores. Figure 1 provides graphical diagnostics of the resulting design.

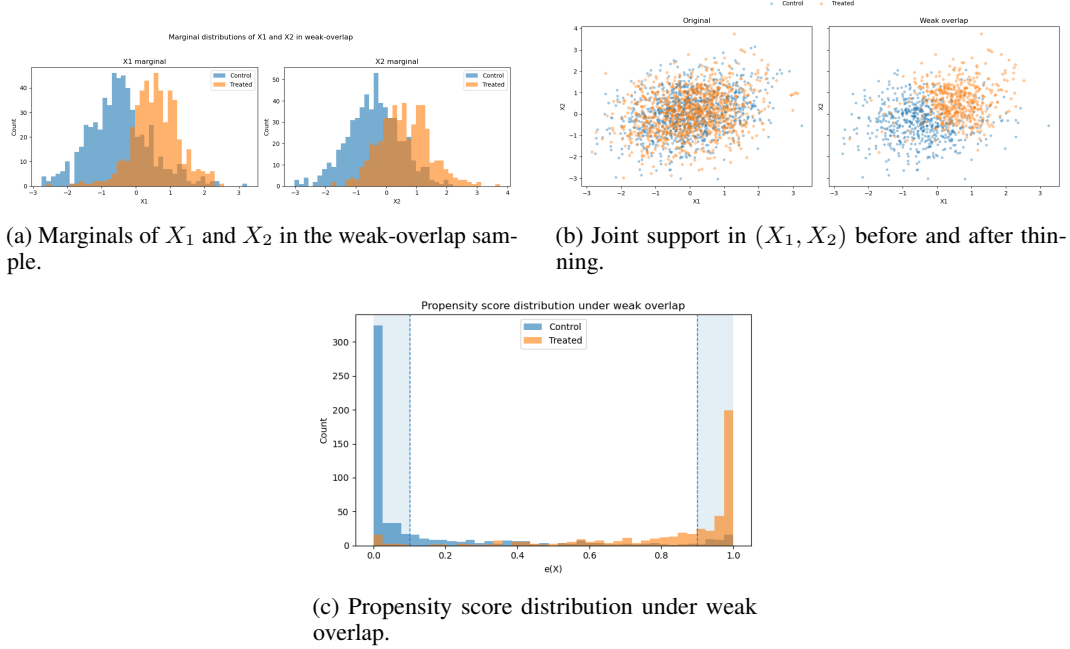


Figure 1: Diagnostics for the induced weak-overlap design. Panel (a) shows that the marginal covariate distributions shift across treatment groups after thinning, although this view can be misleading because the data are 25-dimensional. Panel (b) compares the original and weak-overlap samples in the (X_1, X_2) plane, where the loss of support becomes more visible after thinning. Panel (c) shows the propensity score distribution, where substantial mass accumulates near 0 and 1.

2.2 Method 1: Engression as an Outcome Model

We evaluate whether Engression can improve causal estimation by serving as a flexible outcome model under weak overlap. Our analysis focuses on outcome-regression-based estimators implemented under a T-learner framework, where separate models are fitted for the treated and control groups.

In each simulation, we estimate the conditional outcome functions $\hat{\mu}_t(x)$ using Engression and compute the ATE by plug-in estimation. We further incorporate Engression into the outcome-regression component of the doubly robust AIPW estimator. As baselines, we consider AIPW with Random Forest or OLS outcome models, as well as a direct OLS outcome-regression estimator.

We generate data under a weak-overlap setting (overlap level = 8.0), where more than 70% of observations lie outside the propensity score region (0.1, 0.9). Performance is evaluated using absolute bias and root mean squared error (RMSE) of the estimated ATE.

We also compare S-learner and T-learner formulations and observe a clear difference depending on the overlap regime. Under weak overlap, the T-learner consistently outperforms the S-learner by a substantial margin, while under stronger overlap the S-learner tends to perform better. We report only T-learner results in the main text, with a more detailed comparison provided in Appendix B.

Table 1: Outcome Model Performance (ATE)

Estimator	Mean Abs. Bias	RMSE
Engression (T)	0.034	0.051
AIPW + Engression (T)	0.116	0.131
AIPW + RF (T)	0.173	0.176
AIPW + OLS (T)	0.196	0.196
OLS (T)	1.116	1.116

Note: True ATE: 1.1028. All methods use T-learner outcome modeling. Approximate sample size per simulation is $n \approx 1000$, with 100 simulations. Overlap level = 8.0 (weak).

Table 1 reports the results. The OLS baseline shows extremely biased estimation under weak overlap. Engression as a direct outcome model yields a substantial improvement and achieves the lowest estimation error among all methods, clearly outperforming both classical outcome models and doubly robust estimators. While incorporating Engression into AIPW improves performance relative to standard specifications, it does not match the accuracy of the direct Engression estimator.

Overall, these findings suggest that under weak overlap, directly modeling the full conditional outcome distribution can be more effective than relying on mean-based outcome regression within traditional causal estimators.

2.3 Method 2: Engression for Data Augmentation

In this simulation, we consider a single Engression model for both treatment arms. First, Engression is trained on the weak-overlap sample. Then, for units in extreme propensity-score regions (e.g., $e(x) < 0.10$ or $e(x) > 0.90$), we draw synthetic counterfactual outcomes from $\tilde{Y} \sim \hat{P}_{Y|T=1,X}$ or $\tilde{Y} \sim \hat{P}_{Y|T=0,X}$, respectively. The synthetic observations are then added to the original dataset and evaluated using standard causal estimators with their usual mean outcome models. This yields the following results.

Table 2: Simulation results before and after Engression-based data augmentation. Smaller absolute values for bias and RMSE are highlighted. Lower is better.

Estimator	Bias (No Aug.)	Bias (Aug.)	RMSE (No Aug.)	RMSE (Aug.)
OLS	1.0477	0.4072	1.0529	0.4115
IPW	-0.0981	0.4073	0.1627	0.4116
AIPW-RF	0.0249	0.3863	0.1120	0.3892
Causal Forest	-0.1212	0.3310	0.1471	0.3340
Engression	0.5807	0.3995	0.5810	0.4001

We can try to make sense of this result. Because OLS and the naive Engression estimator are the most directly affected by the lack of covariate support, augmentation may help fill in some of those gaps. In contrast, IPW, AIPW-RF, and Causal Forest rely more heavily on the joint correctness of treatment assignment, nuisance estimation, or local heterogeneity. When the augmented outcomes are model-generated rather than truly observed, these methods can be distorted by the additional noise that the synthetic data introduce.

3 Findings and Discussion

The simulation results show that replacing conventional outcome models with Engression consistently improves estimation performance under weak overlap. In particular, when Engression is used to model the conditional mean outcome, either directly or as the outcome component in AIPW, the resulting estimates show substantially lower bias and variance. This suggests that a more flexible representation of the conditional outcome distribution can partially mitigate extrapolation error in regions where one treatment arm is scarce.

Despite these improvements, the performance of Engression remains sensitive to model quality, as seen for example in the S-learner setting. When the learned conditional distribution fails to adequately approximate the true data-generating process, the resulting estimates can still be biased. This highlights an important limitation: while Engression provides a powerful and flexible tool for addressing weak overlap, it does not eliminate the fundamental challenge of extrapolation, and its success depends critically on the accuracy of the learned distribution.

On the other hand, augmenting the data with counterfactual pairs generated by Engression may not be statistically well justified. By doing so, we may alter the empirical distribution of the observed data, which can negatively affect the ATE estimate. Furthermore, one could argue that Engression should be retrained on the augmented data and then re-evaluated using that reinforced model.

4 Empirical Dataset

4.1 Moderate Overlap: IHDP

We further evaluate our data-augmentation approach using Engression on the Infant Health and Development Program (IHDP) benchmark [Hill, 2011]. This dataset is a standard benchmark for treatment-effect estimation and has moderate overlap. It contains 747 units (139 treated and 608 control) with 25 covariates.

After fitting a propensity model, we identify a weak-overlap region defined by $0.08 \leq \text{PS} \leq 0.25$. Control samples in this region are selected, and counterfactual outcomes $Y_{\text{syn}} \sim P(Y \mid X, T = 1)$ are sampled from the trained Engression model. Using this approach, around 140 synthetic samples are generated, which is approximately the same as the number of originally treated units. The absolute mean standardized mean difference (SMD) across covariates is 0.186 before augmentation and decreases to 0.08, indicating improved covariate balance. For individual-level causal effects, we report Precision in Estimation of Heterogeneous Effect (PEHE),

$$\epsilon_{\text{PEHE}} = \frac{1}{n} \sum_{i=1}^n (\hat{\tau}_i - \tau_i)^2,$$

and RMSE-ITE. For population-level causal effects, we report absolute error in the ATE [Louizos, 2017, Shalit, 2017].

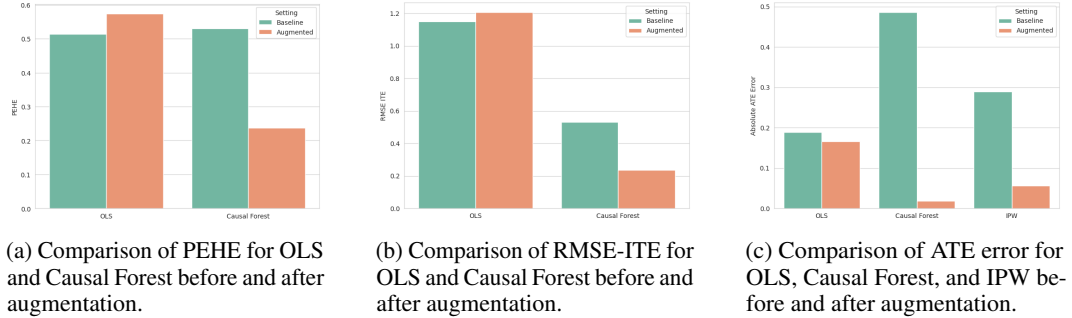
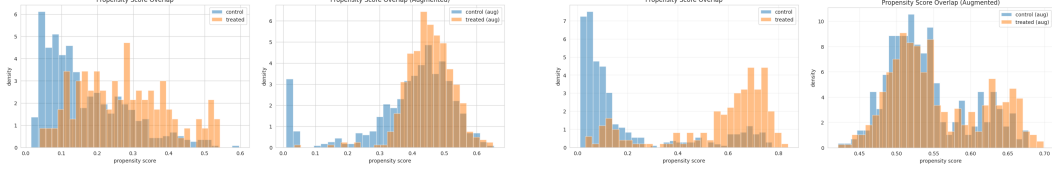


Figure 2: IHDP results.

The results are shown in Figure 2. OLS does not improve for individual-level causal effects because it cannot capture heterogeneous treatment effects. However, it does improve for population-level causal effects. The performance of Causal Forest improves across all three metrics because it can model heterogeneous treatment effects. For IPW, the synthetic data reduce extreme propensity weights, which helps stabilize the estimator.

4.2 Weak Overlap: Jobs

The study of LaLonde [LaLonde, 1986] is another widely used benchmark in causal inference, where job training is the treatment and income is the outcome. The dataset consists of 614 units (185 treated and 429 control) with 8 covariates. Compared with IHDP, this dataset exhibits substantially



(a) Propensity score distributions in the IHDP dataset (moderate overlap).

(b) Propensity score distributions in the LaLonde dataset (weak overlap).

weaker overlap, with mean $|SMD| = 0.513$. Accordingly, the weak-overlap region is larger, defined as $0.1 \leq PS \leq 0.3$, and around 273 synthetic samples are generated.

Here, since the treated group comes from the randomized sample, we report the absolute error of the Average Treatment Effect on the Treated (ATT) [Louizos, 2017, Shalit, 2017]. After augmentation, the mean absolute SMD decreased to 0.089, but the estimation results differ from those of IHDP. For OLS, the estimated ATT decreases from 1634 to 1599. However, Causal Forest and IPW perform worse. In particular, the Causal Forest estimate increases from 1285 before augmentation to 1781 after augmentation.

From our results, we observe that Engression-based augmentation improves covariate balance, but its effectiveness in causal estimation depends on the degree of overlap in the data.

5 Future Work

A direction for future work is to develop a more formal understanding of when distributional regression methods such as Engression are effective for causal inference under weak overlap. In particular, it would be valuable to introduce quantitative measures of discrepancy between the covariate distributions $P(X | T = 1)$ and $P(X | T = 0)$. These measures could serve as diagnostic tools for assessing the difficulty of extrapolation and for guiding the choice of estimation strategy.

In addition, while this work focuses primarily on the ATE, an important extension is to study heterogeneous treatment effects, such as conditional average treatment effects (CATE), and related metrics such as PEHE. Since Engression models the full conditional outcome distribution, it is naturally well suited to capturing treatment-effect heterogeneity, and its benefits may be more pronounced in these settings. At present, we include related metrics only in the empirical data section.

6 Conclusion

In this work, we investigate the use of Engression for causal effect estimation under weak covariate overlap. Our results highlight a clear distinction between two ways of leveraging generative models: outcome modeling and data augmentation.

When used as an outcome model, Engression consistently improves estimation accuracy across simulations. In particular, it achieves substantially lower bias and RMSE for the ATE than competing approaches, especially under weak overlap. This improvement can be attributed to its ability to model the full conditional outcome distribution, which provides a more flexible and stable representation of outcomes and reduces extrapolation error when estimating counterfactual means.

In contrast, Engression-based data augmentation does not consistently improve performance and can even degrade estimation accuracy, particularly for methods such as IPW, AIPW, and Causal Forest. This discrepancy reflects a fundamental difference between the two approaches. Outcome modeling operates conditionally on observed covariates and does not alter the underlying data distribution, while augmentation introduces synthetic samples that modify the joint distribution of (X, T, Y) . When overlap is weak, these synthetic samples may lie far from the true support of the missing counterfactuals, leading to additional noise and estimation bias.

Our empirical results further support this interpretation. In the IHDP dataset, where overlap is moderate, augmentation improves covariate balance and can benefit downstream estimators. However, in the LaLonde dataset, which exhibits stronger overlap violations, augmentation fails to improve

treatment-effect estimation. This suggests that the effectiveness of generative augmentation depends critically on the structure of the covariate space and the degree of overlap between treatment groups.

Overall, our findings indicate that distributional regression is a promising tool for causal inference under limited overlap, particularly when used as an outcome model. However, naive use of generative models for data augmentation can be problematic and requires careful consideration.

These findings motivate the need for formal criteria to characterize when generative approaches such as Engression are effective.

References

- Shen, Y. (2024). Engression: Energy-based neural distributional regression. *Journal of Machine Learning Research*, 25(1), 1–35.
- Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378.
- Crump, R. K., Hotz, V. J., Imbens, G. W., and Mitnik, O. (2009). Dealing with limited overlap in estimation of average treatment effects. *Biometrika*, 96(1), 187–199.
- Hill, J. L. (2011). Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1), 217–240.
- LaLonde, R. J. (1986). Evaluating the econometric evaluations of training programs with experimental data. *The American Economic Review*, 76(4), 604–620.
- Louizos, C., Shalit, U., Mooij, J. M., Sontag, D., Zemel, R., and Welling, M. (2017). Causal effect inference with deep latent-variable models. *Advances in Neural Information Processing Systems*.
- Shalit, U., Johansson, F. D., and Sontag, D. (2017). Estimating individual treatment effect: generalization bounds and algorithms. *Proceedings of the 34th International Conference on Machine Learning*.

7 Appendix

A Full Data-Generating Process

Our simulation study is based on the following data-generating process.

1. **Covariates.** For each unit, draw $X = (X_1, \dots, X_p)^\top \in \mathbb{R}^p$ from a multivariate normal distribution

$$X \sim \mathcal{N}(0, \Sigma), \quad \Sigma_{jk} = 0.3^{|j-k|}.$$

2. **Baseline propensity score and treatment assignment.** Define the treatment propensity score as

$$e(X) = \sigma\left(0.5 \tanh(X_1) + 0.3 X_2 - 0.2 \sum_{j=1}^{\min(5,p)} X_j\right),$$

where $\sigma(u) = 1/(1 + e^{-u})$ is the sigmoid function. Treatment is then assigned as

$$T \mid X \sim \text{Bernoulli}(e(X)).$$

3. **Pre-additive noise.** Draw

$$\eta \sim \mathcal{N}(0, 0.4^2 I_p), \quad Z = X + \eta.$$

4. **Baseline response.** Define the untreated conditional mean as

$$\mu_0(Z) = 2 \sin(Z_1) \cos(Z_2) + 0.5 \tanh\left(\sum_{j=1}^p Z_j\right) + 0.3 \log\left(1 + \sum_{j=1}^p Z_j^2\right).$$

5. **Heterogeneous treatment effect.** Define the conditional treatment effect by

$$\tau(X) = 1.5 + 0.8 \sin(2X_1) - 0.4X_2^2 + 0.2 \cdot \frac{1}{p} \sum_{j=1}^p X_j.$$

6. **Potential outcomes.** Let

$$\mu_1(X, Z) = \mu_0(Z) + \tau(X).$$

Then generate potential outcomes as

$$Y_0 = \mu_0(Z) + \varepsilon_0, \quad Y_1 = \mu_1(X, Z) + \varepsilon_1,$$

where

$$\varepsilon_0, \varepsilon_1 \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 0.5^2).$$

7. **Observed outcome.** The observed response is

$$Y = TY_1 + (1 - T)Y_0.$$

8. **ATE.** The true ATE in the generated super-population is

$$\text{ATE} = \mathbb{E}[\tau(X)].$$

9. **Weak-overlap construction.** Starting from the generated super-population, we induce weak overlap through a covariate-dependent thinning procedure. Specifically, define

$$e_{\text{weak}}(X) = \sigma(\gamma(0.6X_1 + 0.4X_2 - 0.3X_1^2)),$$

where larger values of γ produce sharper separation. Observations are then retained with probabilities

$$r(X, T) = \begin{cases} 1 - e_{\text{weak}}(X), & T = 0, \\ e_{\text{weak}}(X), & T = 1, \end{cases}$$

up to a small clip to prevent complete removal of one tail. Thus, treated units are preferentially retained in regions with high $e_{\text{weak}}(X)$, while control units are preferentially retained in regions with low $e_{\text{weak}}(X)$. This creates sparse counterfactual support in the tails and yields a weak-overlap sample overall.

B S-Learner and T-Learner Comparison

In this section, we compare the S-learner and T-learner frameworks under weak overlap. Here, the S-learner models both treatment arms with a single model, while the T-learner fits separate models for the treated and control groups.

Table 3 shows that the T-learner consistently outperforms the S-learner across nearly all estimators. In particular, the performance gap is especially large for Engression-based methods: the direct Engression estimator achieves substantially lower bias and RMSE under the T-learner, while its S-learner counterpart exhibits noticeably degraded performance. A similar pattern is observed for AIPW with Engression, where the S-learner version also performs considerably worse than the T-learner version.

This difference may be attributed to the difficulty of modeling both treatment arms with a single conditional model under weak overlap. In regions of the covariate space where one treatment group is underrepresented, the S-learner must rely heavily on extrapolation, which leads to unstable estimates. In contrast, the T-learner fits separate models for treated and control groups, allowing it to better adapt to imbalance in the data and reduce extrapolation bias.

We also explore different covariate dimensionalities and overlap levels to examine the performance differences between T-learners and S-learners. Figure 4 shows that when overlap is relatively weak, the RMSE of the two approaches becomes comparable. As the dimensionality of the covariates increases, S-learners can even outperform T-learners.

This pattern suggests that while T-learners are generally preferred for capturing heterogeneous treatment effects, their advantage diminishes in high-dimensional settings where splitting the data by treatment leads to increased estimation variance. In contrast, S-learners benefit from pooling information across treatment groups, which can improve stability when data become sparse in each subgroup. Overall, the relative performance of S- and T-learners depends on the interplay between overlap and dimensionality, which sheds some light on the settings in which each method should be used.

Overall, these results show that under weak overlap, the T-learner provides a more robust framework for outcome modeling, particularly when combined with flexible distributional methods such as Engression. This motivates our focus on T-learner results in the main text.

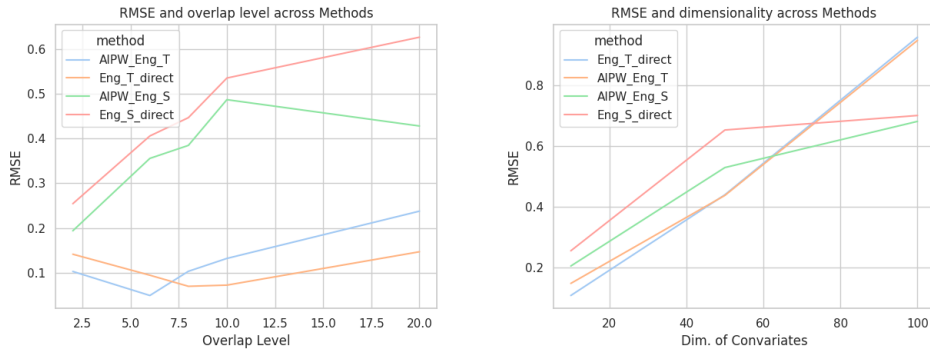


Figure 4: RMSE as a function of overlap and dimensionality.

Table 3: Outcome model performance (T- and S-learners)

Estimator	Mean Abs. Bias	RMSE
Engression (T)	0.034	0.051
AIPW + Engression (T)	0.116	0.131
AIPW + RF (T)	0.173	0.176
AIPW + RF (S)	0.182	0.184
AIPW + OLS (S)	0.193	0.193
AIPW + OLS (T)	0.196	0.196
AIPW + Ridge (S)	0.199	0.199
AIPW + Engression (S)	0.385	0.386
Engression (S)	0.436	0.437
OLS (S)	1.053	1.053
OLS (T)	1.116	1.116

Note: True ATE: 1.1028. (T) and (S) denote **T-learner** and **S-learner**, respectively. Approximate sample size per simulation is $n \approx 1000$, with 100 simulations. Overlap level = 8.0 (weak).